User Manual

Smart Hostel Outpass Management System Using NLP

For students and hostel wardens. Screenshots are from the running application.

1. Installation (Windows 11, PowerShell)

Install Python 3.10 or newer from python.org and tick **Add python.exe to PATH**. Then run these commands in PowerShell:

```
Expand-Archive -Path "$HOME\Downloads\outpass_system.zip" -DestinationPath
"$HOME\Desktop" -Force
cd "$HOME\Desktop\outpass_system"
python -m venv venv
.\venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
python init_db.py
python train_model.py
python seed_demo_requests.py
python app.py
```

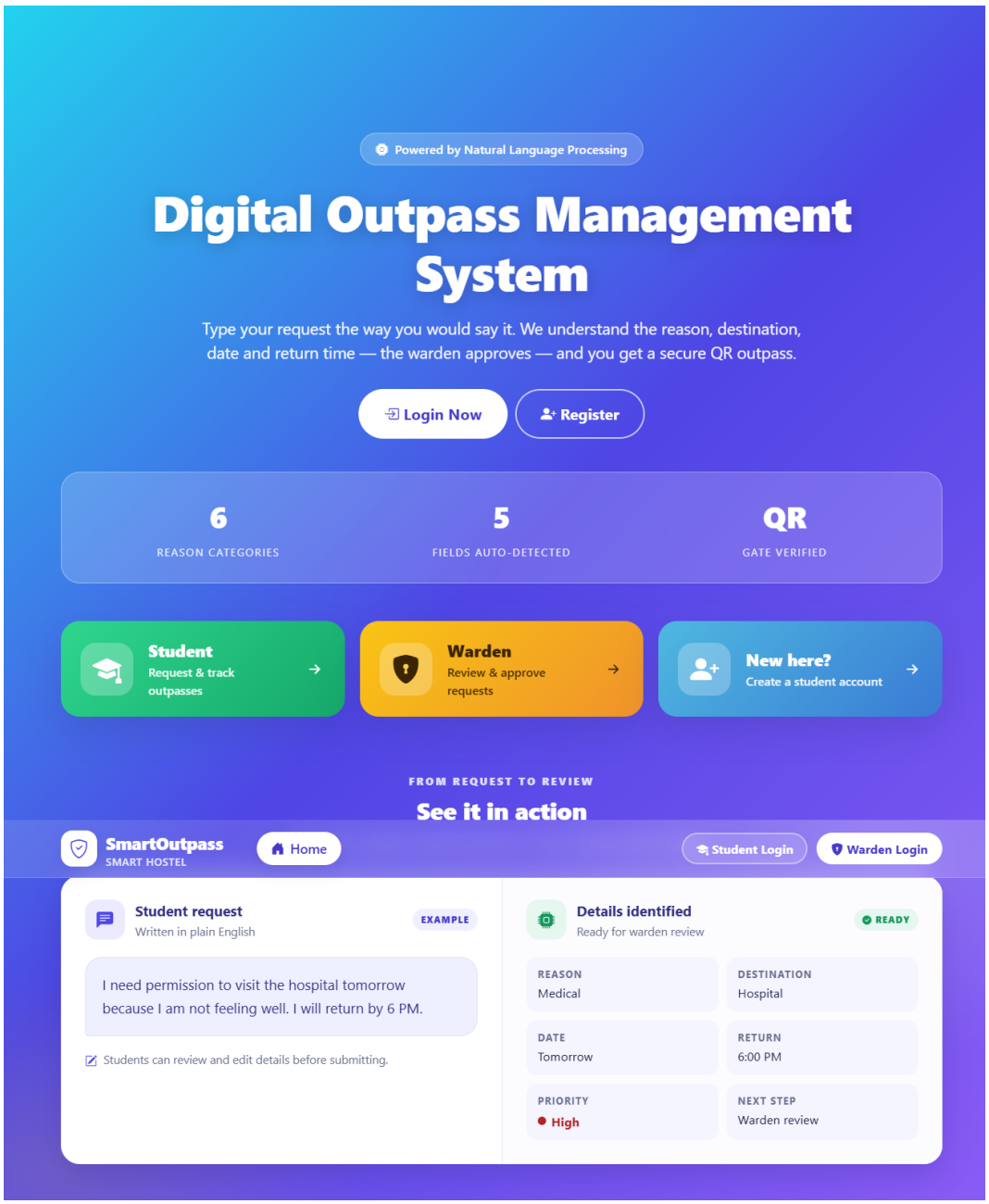
If activation is blocked, run `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned` once. Open **http://127.0.0.1:5000** in Chrome or Edge. Press Ctrl+C in PowerShell to stop the server.

Demo logins

Role	Login	Password
Warden	warden	Warden@123
Student	21CS001	Student@123
Student	21EC014	Student@123

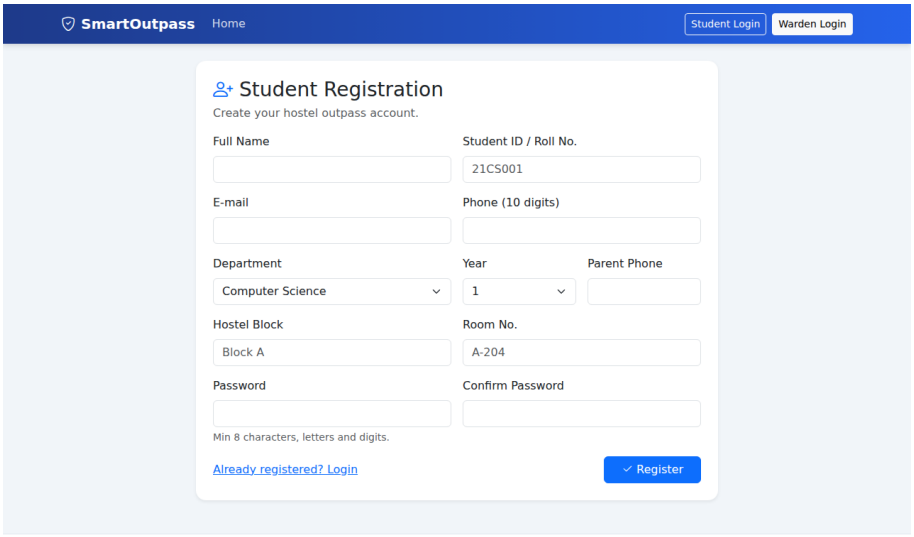
2. Student guide

2.1 Home, registration and login

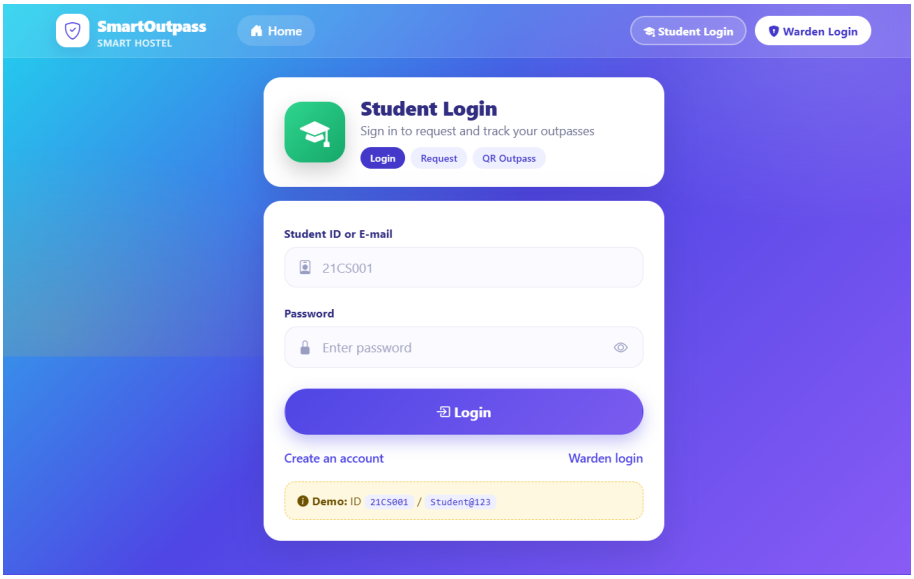


Home page

Click **Register**, fill in your details (10-digit phone, password of 8+ characters with letters and digits) and submit. Then log in with your Student ID or e-mail.

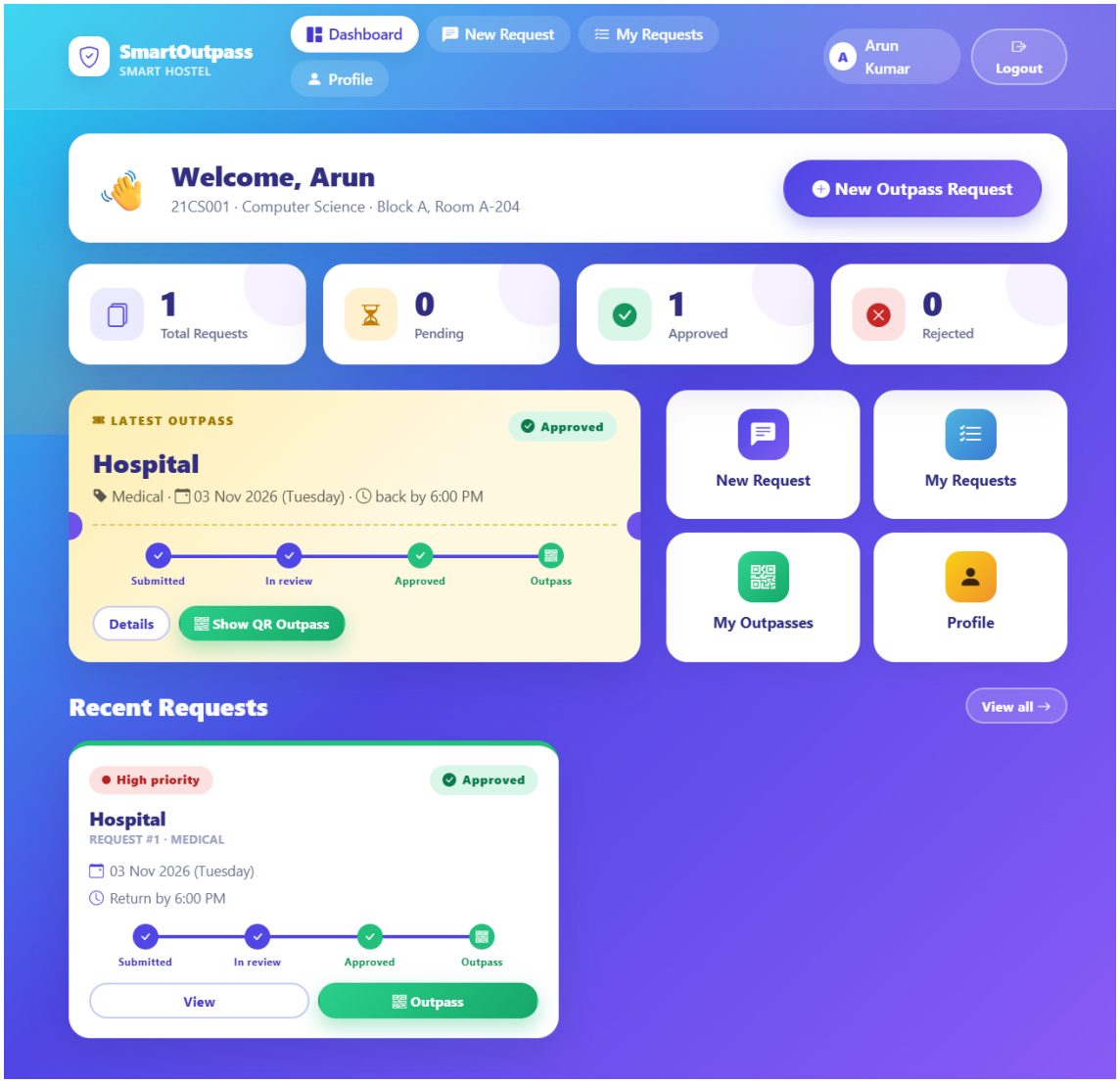


Registration page

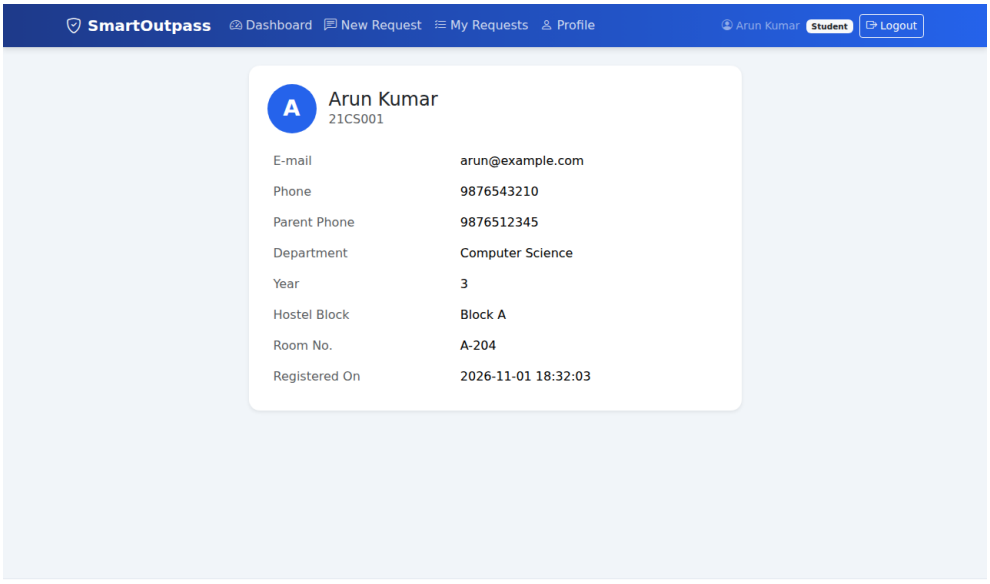


Student login

2.2 Dashboard and profile



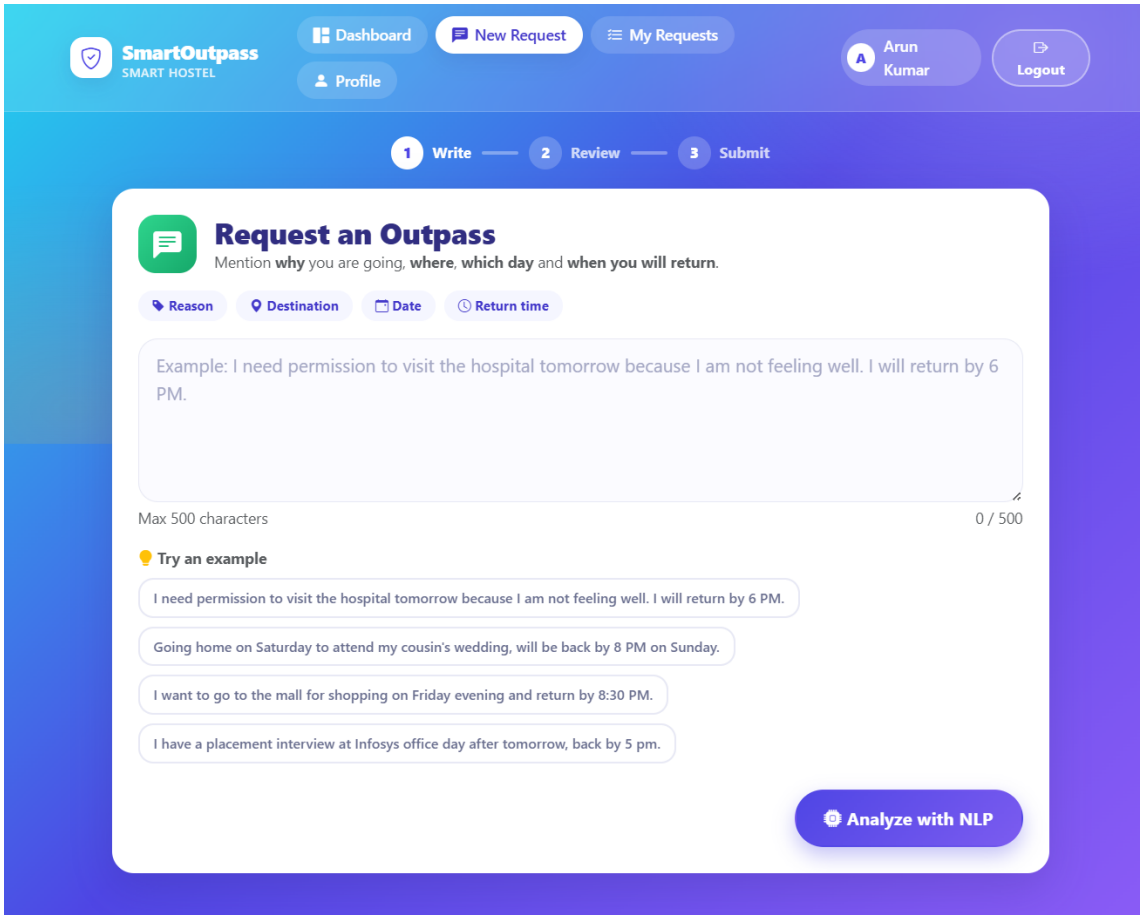
Student dashboard with request counts



Profile page

2.3 Submit a request in natural language

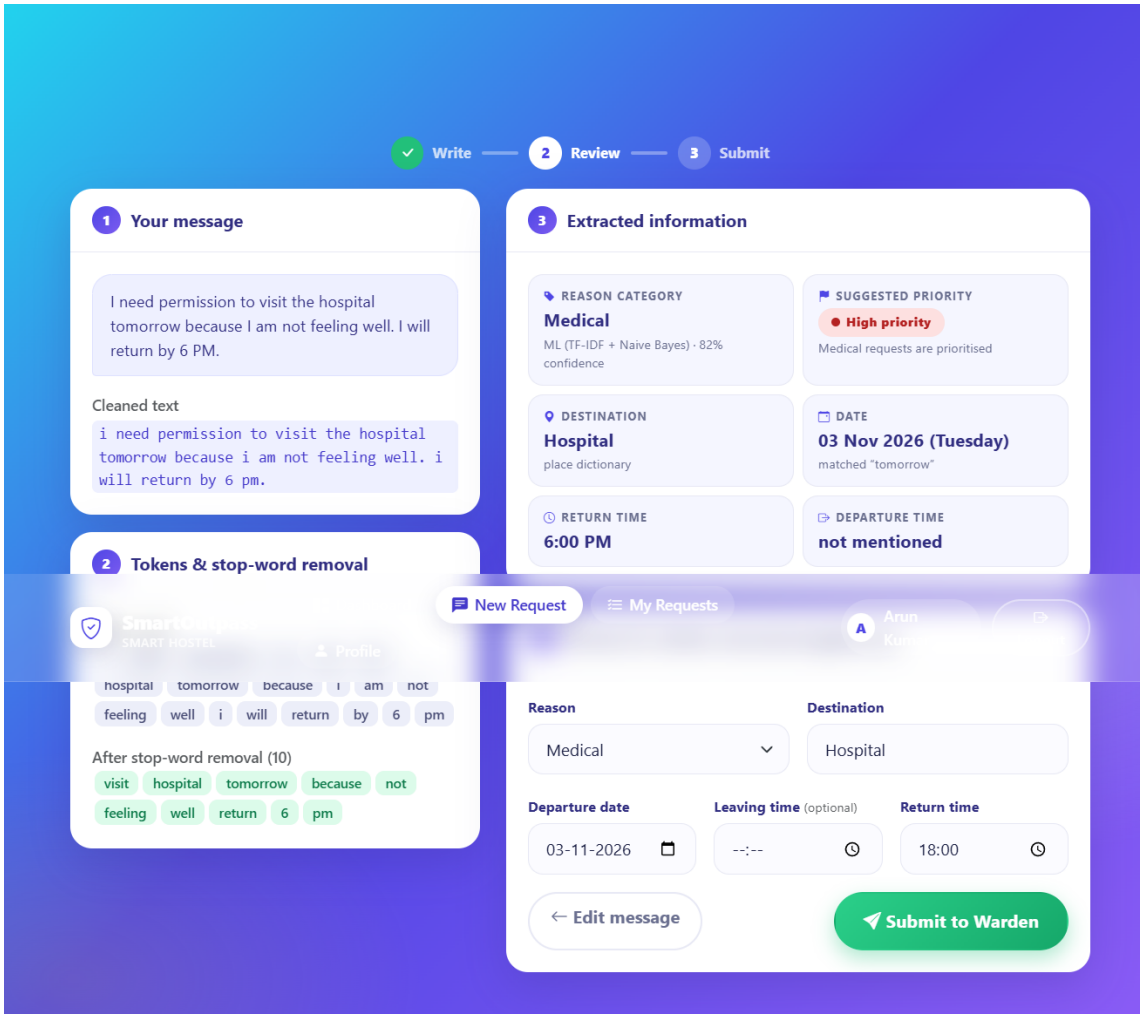
Click **New Request** and type your message, for example: *I need permission to visit the hospital tomorrow because I am not feeling well. I will return by 6 PM.* Include the reason, place, day and return time.



Request form with example buttons

2.4 Check the NLP analysis

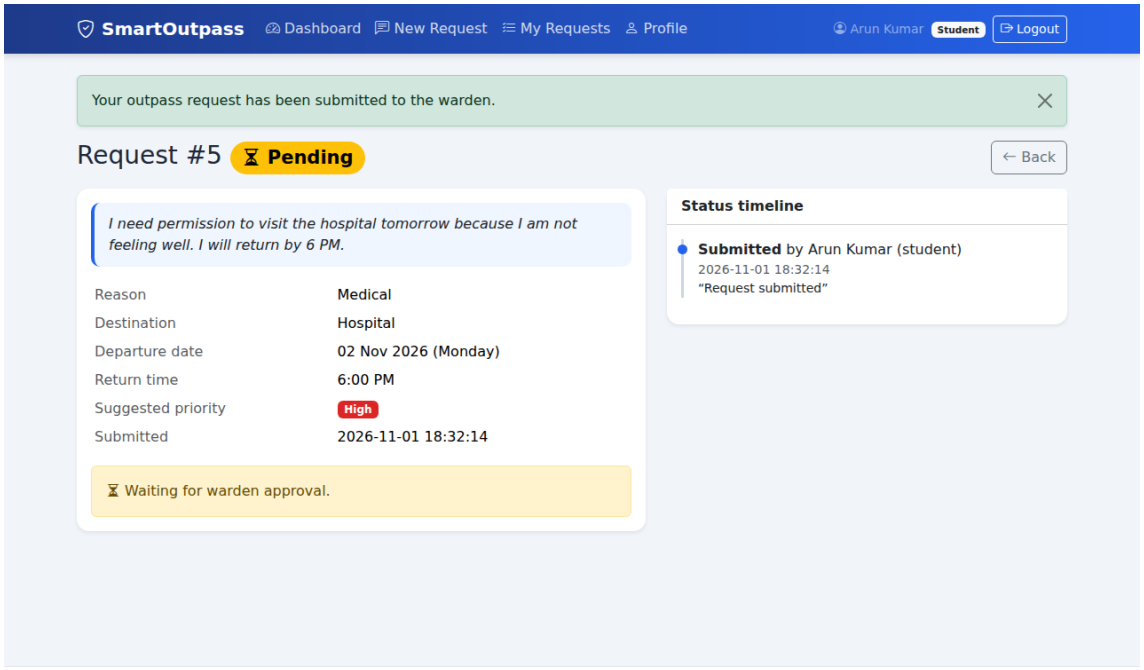
Press **Analyze with NLP**. The page shows the tokens, the detected reason, destination, date, return time and suggested priority. If information is missing the fields are highlighted - fill them in. You may correct any value, then click **Submit to Warden**.



NLP analysis and confirmation page

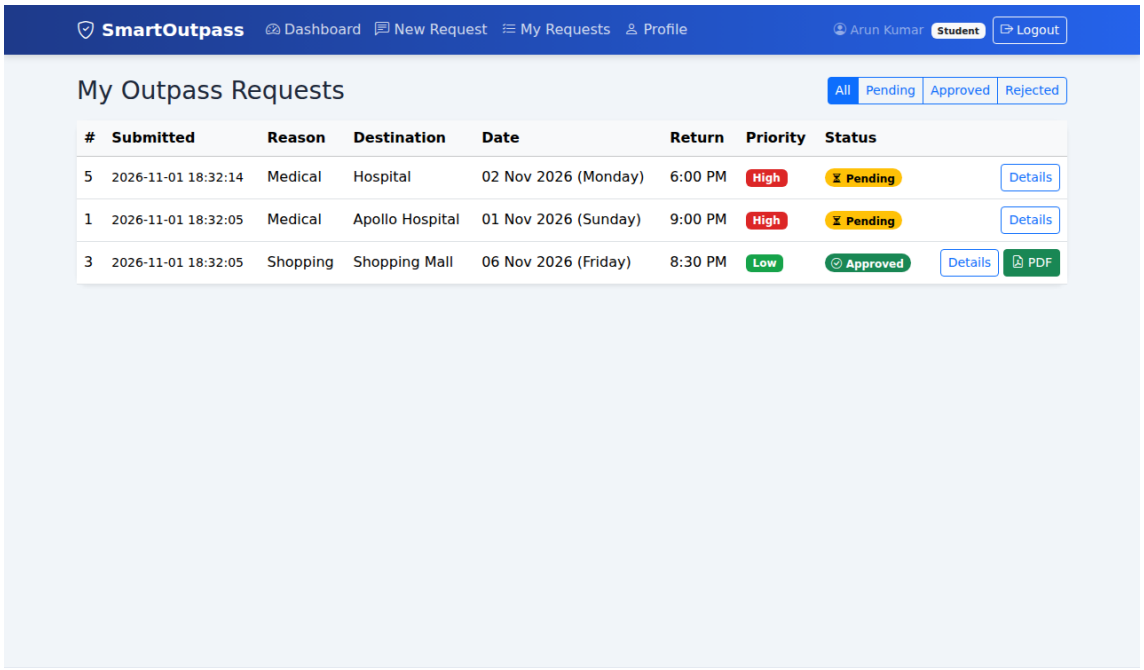
2.5 Track status and download the outpass

The status page shows a timeline. When the warden approves, **View Digital Outpass** and **Download PDF** appear. Show the PDF (with its QR code) at the gate.



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Request status page



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My requests list



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Digital outpass page

Sri Vidya Institute of Engineering & Technology

Central Boys & Girls Hostel

DIGITAL HOSTEL OUTPASS

Outpass ID: OP20261101-0002 **STATUS: APPROVED**

Student Name	Priya Sharma
Student ID	21EC014
Department / Year	Electronics / Year 2
Hostel / Room	Block B, Room B-110
Reason for Leaving	Family Function
Destination	Home
Departure Date	04 Nov 2026 (Wednesday)
Expected Return Time	8:00 PM
Priority	Medium
Approved By	Dr. R. Meenakshi (Chief Hostel Warden)
Approved On	2026-11-01 18:32:19
Warden Remarks	Approved. Return before 6 PM.

Dr. R. Meenakshi
Chief Hostel Warden

Scan to verify this outpass

Instructions

1. Show this outpass (printed or on phone) to the security guard at the gate.
2. Return to the hostel before the expected return time mentioned above.
3. This outpass is valid only for the date and destination stated.
4. Misuse or late return may lead to disciplinary action.

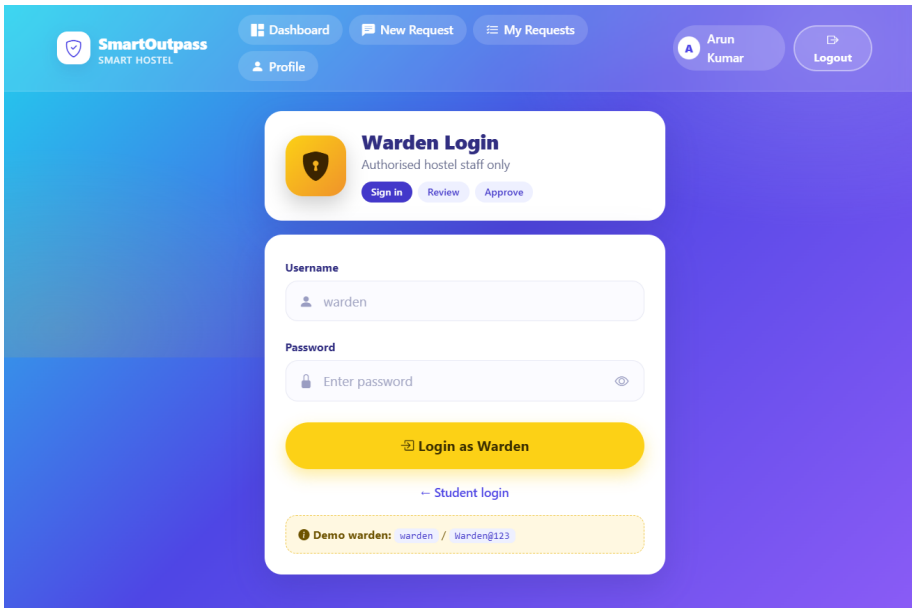
Computer-generated document. Verification link: http://localhost/verify/OP20261101-0002/1b89d6c3fb895f72

Downloaded PDF outpass

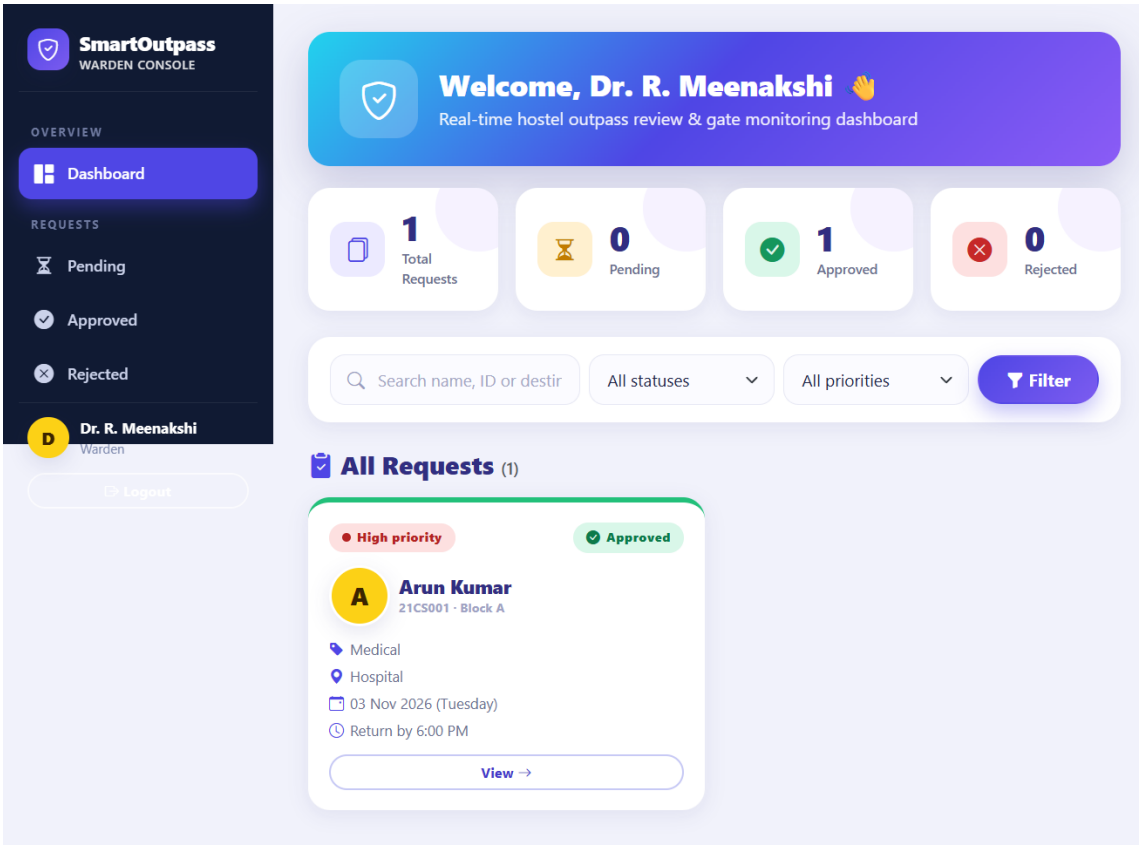
3. Warden guide

3.1 Login and dashboard

Use **Warden Login** (warden / Warden@123). The dashboard lists all requests, pending first and sorted by priority. Use the search box and filters to find requests.



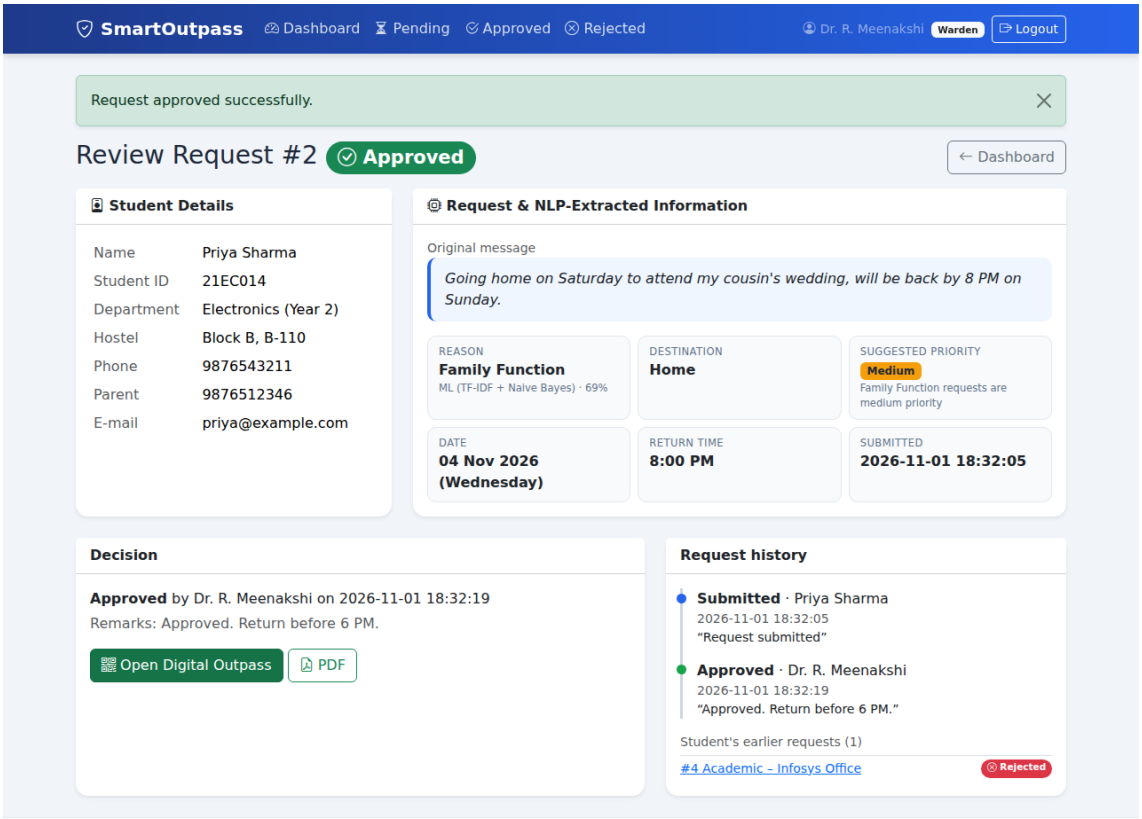
Warden login



Warden dashboard

3.2 Review, approve or reject

Click **Review** to see the student's details, the original message, the NLP-extracted information, the suggested priority and the history. Enter remarks and click **Approve** or **Reject** (a remark is mandatory for rejection). A decision cannot be changed afterwards.



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Request after approval



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Digital outpass opened by the warden

4. Troubleshooting

Problem	Solution
'python' is not recognised	Reinstall Python and tick 'Add to PATH', then reopen PowerShell.
Script activation blocked	Run Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
Port 5000 already in use	Close the other program or change the port in the last line of app.py.
Start with clean data	python init_db.py --reset, then python seed_demo_requests.py
Session expired / 400 error	Reload the page and try again (CSRF token refresh).